

EE144 Lab #3

Localization using Unscented Kalman Filter

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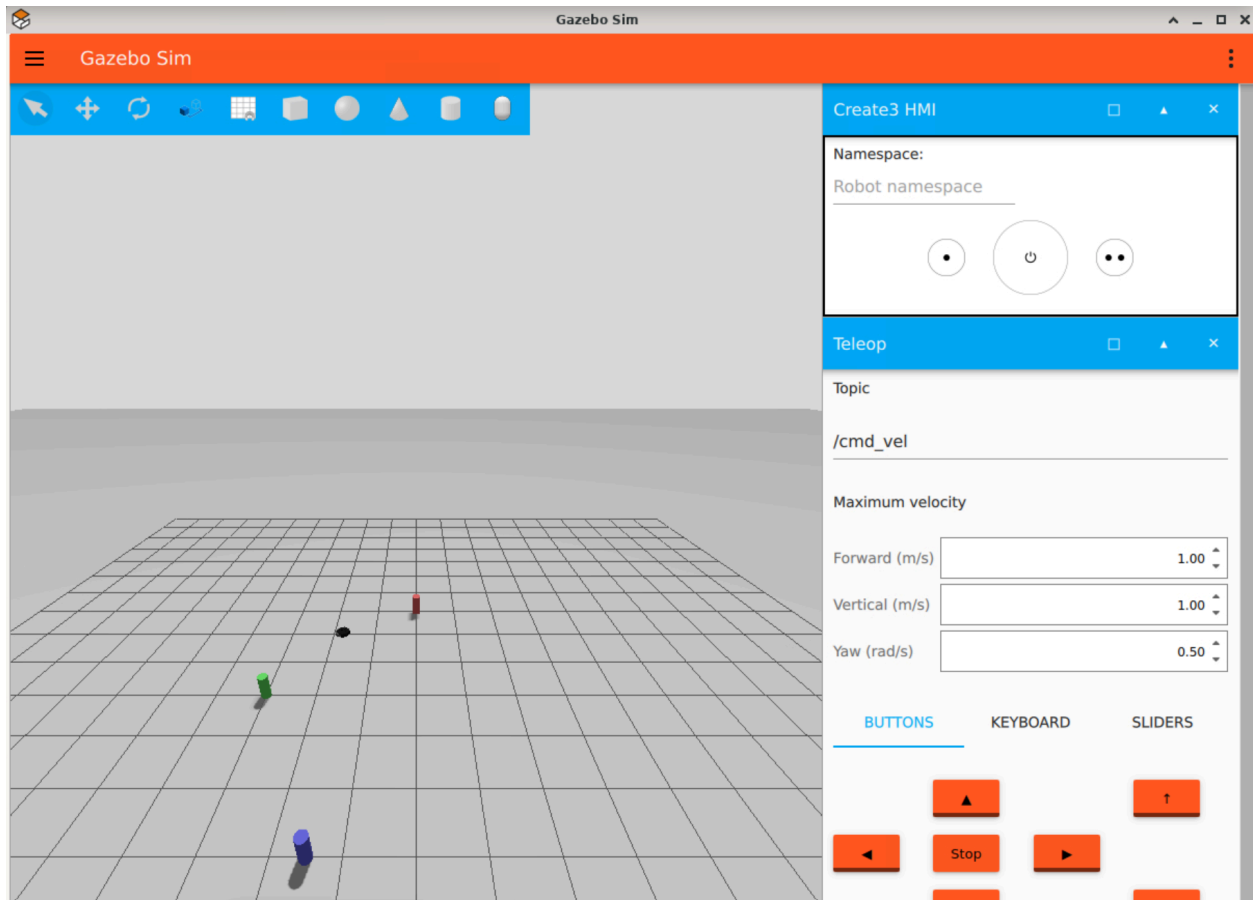
Description

This lab focuses on the Unscented Kalman Filter. It takes the base idea of the Kalman filter of “where do I think I moved to” vs “where I actually moved to” and takes into account the noise of the sensors. Since Kalman filters need linearized systems, the unscented portion creates sample points around the robot’s state, pushes them through nonlinear equations, then reconstructs new Gaussian distributions for the Kalman filter to read through and create estimates. This is particularly powerful because by generating sigma points, the filter is utilizing its nonlinear estimate state to provide the estimated updates it needs to perform.

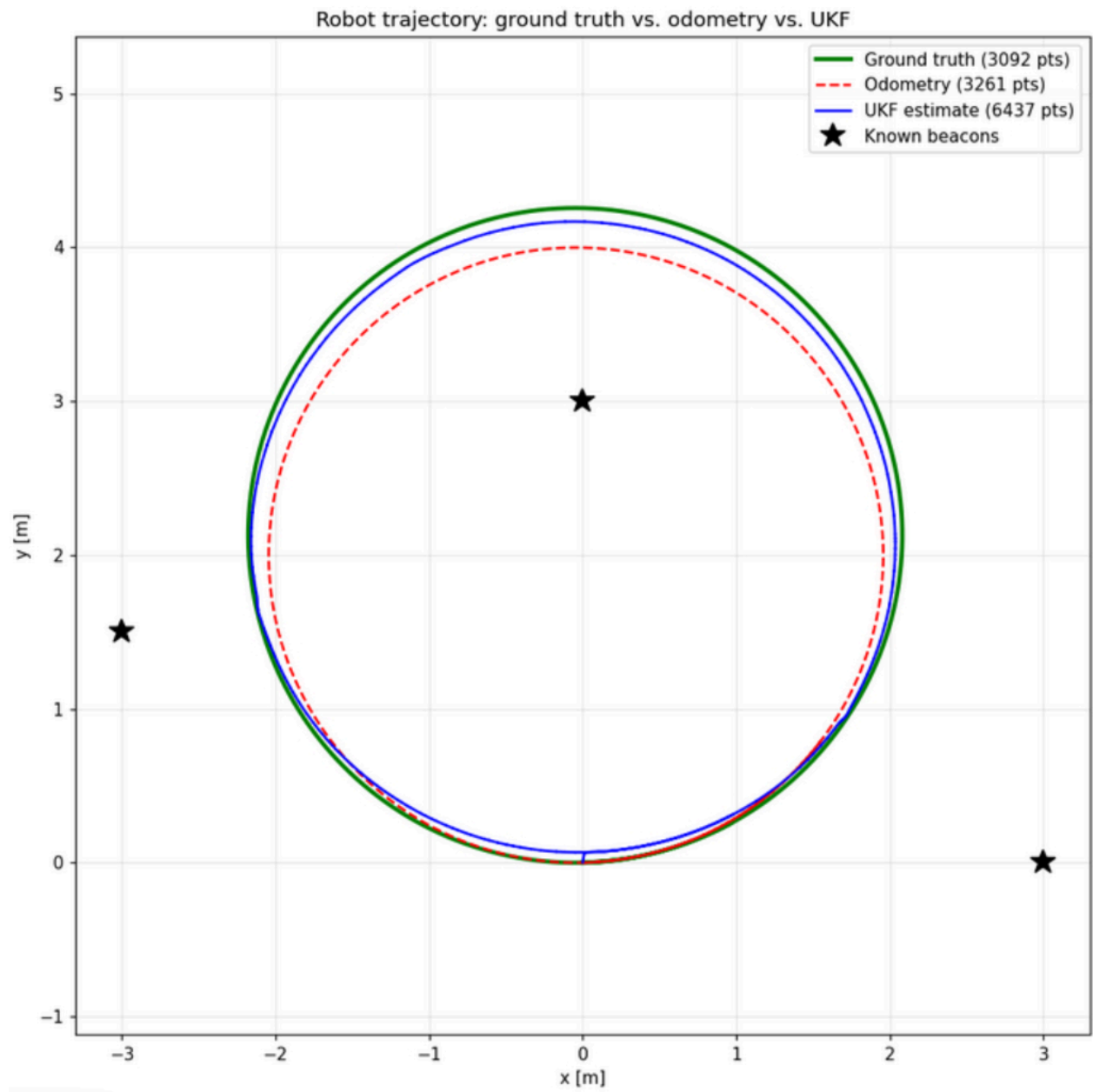
The Unscented Kalman Filter differs from the Extended Kalman Filter in how they interact with nonlinear equations. EKF’s take their nonlinear equations and linearize them with the Jacobian matrices. Once the data has been linearized, it applies normal Kalman math and solves the estimates that way. Unscented Kalman Filters generate sigma points at the robot’s estimated state and push these values through nonlinear equations. Once it finishes this process, it creates new Gaussian equations which the Kalman filter utilizes to provide its estimates. The overall idea difference is that the EKF estimates the changes through tangents while the UKF analyzes the changes through nonlinear equations.

Results

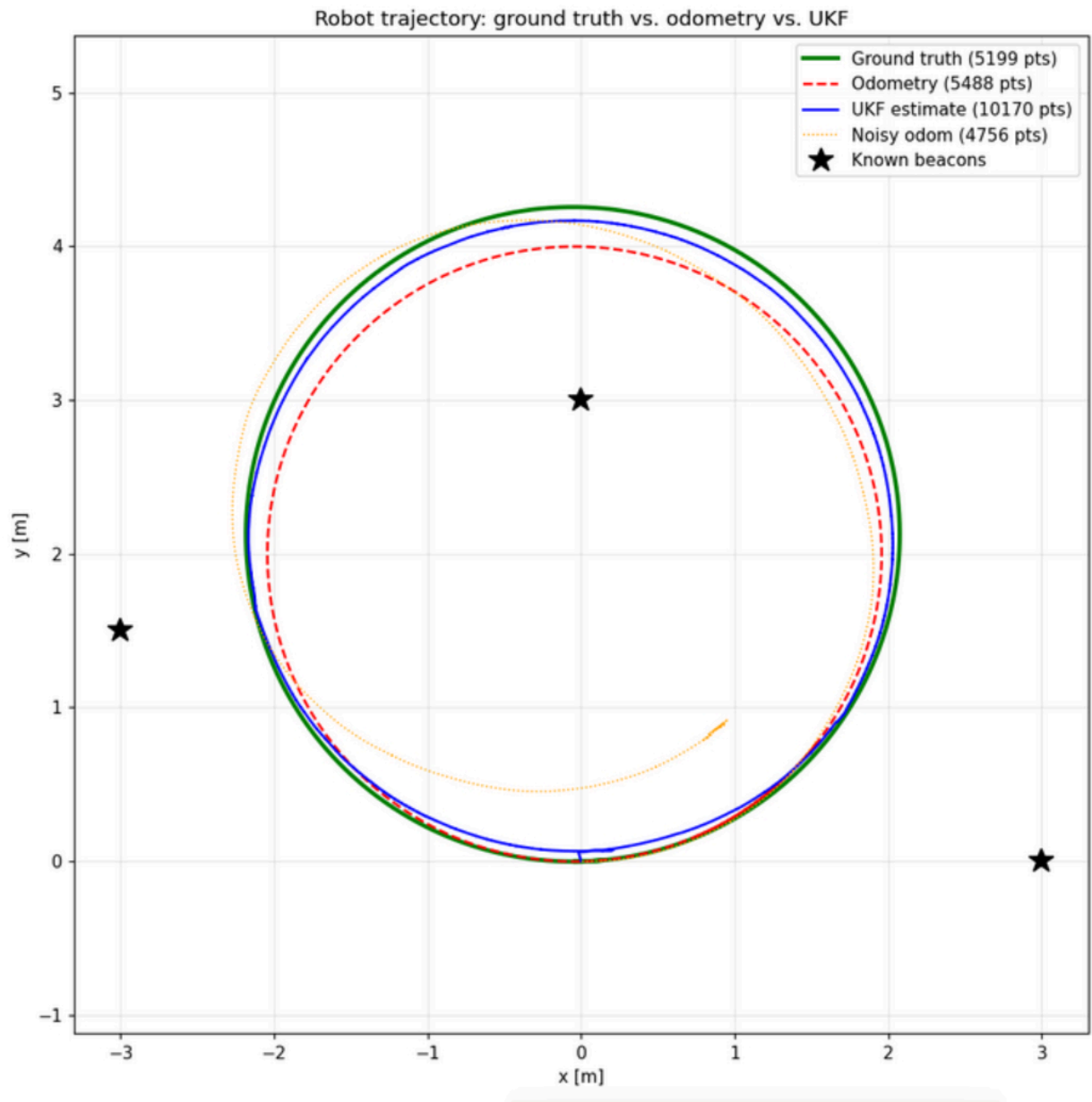
Robot in Motion:



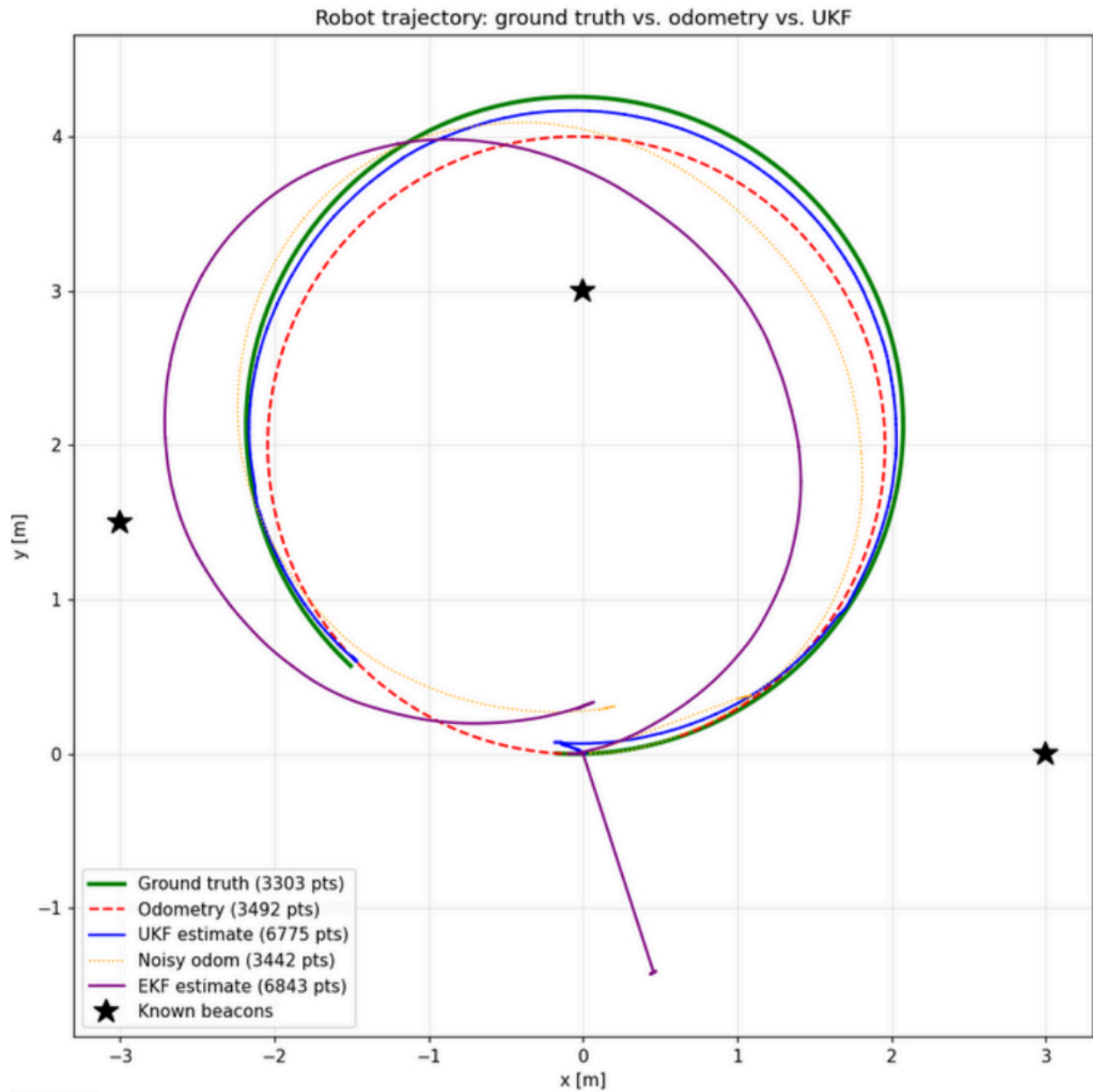
3-Line UKF Plot:



4-Line UKF Plot:



5-Line UKF + EKF Plot:



Discussion

The UKF did track ground truth better than odometry. Looking at the plot above, the blue line barely strayed from ground truth. The UKF was able to correct itself using the LiDAR landmark measurements. When we changed the odometry subscription to noisy, the UKF estimate became less stable and the shape of the UKF increased. This is because the UKF became less confident, however the UKF estimate was still closer to ground truth than base Odometry. Under noisy conditions, the UKF generally performed better than the EKF because of the sampling technique and analysis through the Sigma points. Instead of relying on Jacobian linearization approximations, the Kalman filter utilizes the nonlinear equations it creates to

reconstruct the Gaussian matrices. From there, the values will be more reliable because we are not using tangent lines to approximate the movement. The covariance filter behaves similarly to the EKF. When the UKF reaches uncertain values, the matrix increases and when the measurement successfully updates, the covariance matrix shrinks. The UKF handled nonlinear uncertainty more accurately, specifically when the robot needed to conduct turns and measure state estimates.